

Touch-First Research Tools for Field Work

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Abstract

Researchers in the field rarely sit at a desk. Notes are taken on phones, data is captured between conversations, and prototypes are tested in hallways. This paper reports on a usability study of a touch-first capture tool designed for student research groups at MGM Laboratory.

1. Introduction

Field research imposes constraints that desk software ignores: one-handed input, intermittent connectivity, and short attention windows. We studied three student research groups over two full cycles of data collection and extracted four constraints that drove the design of a capture-first prototype.

2. Method

Eight participants used the prototype over two weeks. Every entry carried a timestamp and a source, and classification was moved to a review stage after the field session. We measured capture latency, entry completeness, and self-reported flow disruption.

3. Results

Capture latency dropped by 38 percent compared with the participants' previous toolchain. Entry completeness improved most when classification was deferred to review. The single large capture button survived every iteration of the design.

4. Conclusion

Touch-first research tools should optimise for capture speed and defer classification. Testing in the same hallways where the tool is used surfaces interaction problems within minutes instead of days.